

题 1

考虑 $\mathbb{C}$ 中 m 个互异点的补集 $X = \mathbb{C} \setminus \{p_1, \dots, p_m\}$ 。求: $\pi_1(X)$ (不声明基点时都指同构类, 下同)。

Solution: Note that X deformation retracts to $\mathbb{S}^1 \vee \mathbb{S}^1 \vee \dots \vee \mathbb{S}^1$ (m circles). Hence $\pi_1(X) \cong F_m$ which is the free group of rank m .

题 2

乘积空间 $X \times Y$, 关于基点 (x_0, y_0) 。求证: $\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$ 。

Proof: Recall $f: Z \rightarrow X \times Y, f(z) = (g(z), h(z))$ is continuous iff g, h are both continuous. Hence a loop in $X \times Y$ corresponds to loops in X and Y , and likewise a homotopy in $X \times Y$ corresponds to homotopies in X and Y . Therefore we have a bijection $\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0), [f] \mapsto ([g], [h])$, and clearly it is a well-defined isomorphism of groups.

题 3

如果 A 是 X 的子空间, 说 A 是 X 的收缩核 (retract), 意思是存在 $r: X \rightarrow A$ 使得 $r(x) = x$ 对任意 $x \in A$ 成立。举例说明: 收缩核不一定是形变收缩核。

Solution: Consider any non-contractible space like $\mathbb{S}^1$, then there is a retraction to a single point, but no (weak) deformation retraction.

题 4

求证: 边界不是 Möbius 带的收缩核。

Proof: If M retracts to the boundary ∂M , then the inclusion $\iota: \partial M \rightarrow M$ induces a homomorphism $\iota_*: \pi_1(\partial M) \rightarrow \pi_1(M)$. Suppose $\pi_1(\partial M)$ is generated by the circle a , and $\pi_1(M)$ by the circle b , then $\iota_*(a) = 2b$, since by composing the homotopy $M \simeq S^1$ (S^1 is the center strip of M) the image of a winds around S^1 twice. However, the retraction $r: M \rightarrow \partial M$ induces a homomorphism r_* such that $r_* \circ \iota_* = id$, a contradiction.

(Or to avoid discussion on base points, just replace π_1 with H_1 .)

题 5

将 $S^1 \times D^2$ 和 $D^2 \times S^1$ 的边界用同胚 $\phi: S^1 \times S^1 \rightarrow S^1 \times S^1$ 粘合, 其中 S^1 看作复平面单位圆周, $\phi(z, w) = (z^2 w, zw)$ 。所得的空间记为 X 。求: $\pi_1(X)$ 。

Solution: $\pi_1(\mathbb{S}^1 \times \mathbb{D}^2) = \pi_1(\mathbb{S}^1) \times \pi_1(\mathbb{D}^2) = \mathbb{Z}$ is generated by the path a on $\mathbb{S}^1 \times \{1\}$, and likewise $\pi_1(\mathbb{D}^2 \times \mathbb{S}^1) = \mathbb{Z}$ is generated by b on $\{1\} \times \mathbb{S}^1$. Note that a is mapped to two paths $2c$ and b , where c is the circle $\mathbb{S}^1 \times \{1\}$ on $D^2 \times S^1$, so a is nullhomotopic, and a, b are homotopic. Therefore $\pi_1(X) = 0$ is the trivial group.

题 6

考虑 $\mathbb{C} \times \mathbb{C} \times \mathbb{C}$ 的子集 $X = T_1 \cup T_2 \cup T_3$, 其中 $T_1 = \{1\} \times S^1 \times S^1, T_2 = S^1 \times \{1\} \times S^1, T_3 = S^1 \times S^1 \times \{1\}$. 求: $\pi_1(X)$.

Solution: Note that $\pi_1(T_1) = \langle 1 \rangle \times \langle b \rangle \times \langle c \rangle \cong \mathbb{Z}^2$ and likewise $\pi_1(T_i) \cong \mathbb{Z}^2$. Also, $\pi_1(T_i \cap T_j) \cong \mathbb{Z}$ is path-connected and identify the generators of $\pi_1(T_i)$ with each other. Since $T_1 \cap T_2 \cap T_3 = \{1\}$ is connected, and we can take small open neighborhoods of T_i in X , by van Kampen's theorem $\pi_1(X) \cong \mathbb{Z}^3$.

Or note that X is the 2-skeleton of $T^3 = S^1 \times S^1 \times S^1$, so $\pi_1(X) = \pi_1(T^3) = \mathbb{Z}^3$.

题 7

从 $[-2, 2] \times [-2, 2] \times [-2, 2]$ 中去掉三条线段 $I_1 = [-2, 2] \times \{1\} \times \{-1\}, I_2 = \{-1\} \times [-2, 2] \times \{1\}, I_3 = \{1\} \times \{-1\} \times [-2, 2]$, 所得空间记为 X . 求: $\pi_1(X)$.

Solution: Clearly X deformation retracts to $S^1 \vee S^1 \vee S^1$, by removing a regular neighborhood of the three segments, taking point $(0, 0, 0)$ and drawing three circles around each I_i . Hence $\pi_1(X) \cong \mathbb{Z} * \mathbb{Z} * \mathbb{Z}$.

题 8

设 $f: (X, x_0) \rightarrow (Y, y_0)$ 是保基点的连续映射, X 道路连通. 求证: 映射锥的基本群 $\pi_1(C(f), \text{pt})$ 同构于 $\pi_1(Y, y_0)$ 商去 $f_{\#}\pi_1(X, x_0)$ 的正规闭包, 其中 pt 记锥点 (即 $X \times \{0\}$ 坍缩成的那个点).

Solution: Note that CX is simply connected, and $CX \cap Y = f(X)$ is path-connected. Take open neighborhoods U, V of CX and Y , such that U is contractible and $V \simeq Y$, then by van Kampen's theorem, $\pi_1(C(f), \text{pt}) \cong \pi_1(C(f), y_0) = \pi_1(Y, y_0) / \langle\langle f_{\#}\pi_1(X, x_0) \rangle\rangle$.

注记 假设 A 是 X 的子空间, 道路连通且包含基点 x_0 , 则相对拓扑锥 $C(X, A)$ 就是指含入映射 $i_A: A \rightarrow X$ 的映射锥 $C(i_A)$. 由题 8, 可以通过 A 和 X 的基本群表达 $C(X, A)$ 关于锥点的基本群. 但是, 一般不能直接用它来计算 X/A 的基本群, 因为 $C(X, A) \simeq X/A$ 需要更多条件. 如果 A 在 X 中邻域足够好, 比如 A 是流形 X 的正则子流形, 就可以保证 $C(X, A) \simeq X/A$. 代数拓扑中更常用的条件是 (X, A) 是 CW 复形偶, 以后会谈到.

题 9

把 S^1 看作复平面单位圆周. 对任何 $m \in \mathbb{Z}$, 用 $p_m(z) = z^m$ 定义映射 $p_m: S^1 \rightarrow S^1$. 求: $\pi_1(C(p_m))$.

Solution: The polygonal presentation of $C(p_m)$ is a^m , so its fundamental group is $\langle a | a^m = 1 \rangle = \mathbb{Z}/m\mathbb{Z}$.

题 10

对于带基点的拓扑空间 (X, x_0) , 基点处闭道路的定端同伦类 γ 诱导基本群的自同构 $\gamma_{\#}$. 求证: 这样, $\gamma \mapsto \gamma_{\#}$ 给出一个群同态 $\pi_1(X, x_0) \rightarrow \text{Aut}(\pi_1(X, x_0))$.

Proof: Note that $\gamma_*(f) = \gamma \cdot f \cdot \gamma^{-1}$, so it just gives the conjugate automorphism. Clearly $(\gamma\delta)_* = \gamma_* \circ \delta_*$, so it is a group homomorphism.

题 11

问: 题 10 的同态是否总为单同态?

Solution: No. The kernel will be the center $C(G)$, which may be non-trivial for abelian groups like $G = \pi_1(S^1) = \mathbb{Z}$.

题 12

问：题 10 的同态是否总为满同态？

Solution: No. If G is abelian, then the image is zero, but for example if $G = \pi_1(S^1 \times S^1) = \mathbb{Z}^2$ then $\text{Aut}(G) = \text{GL}_2(\mathbb{Z})$ is non-trivial. Hence there is no epimorphism $G \rightarrow \text{Aut}(G)$.